Ghosting Isn't Cowardice. It's Math.

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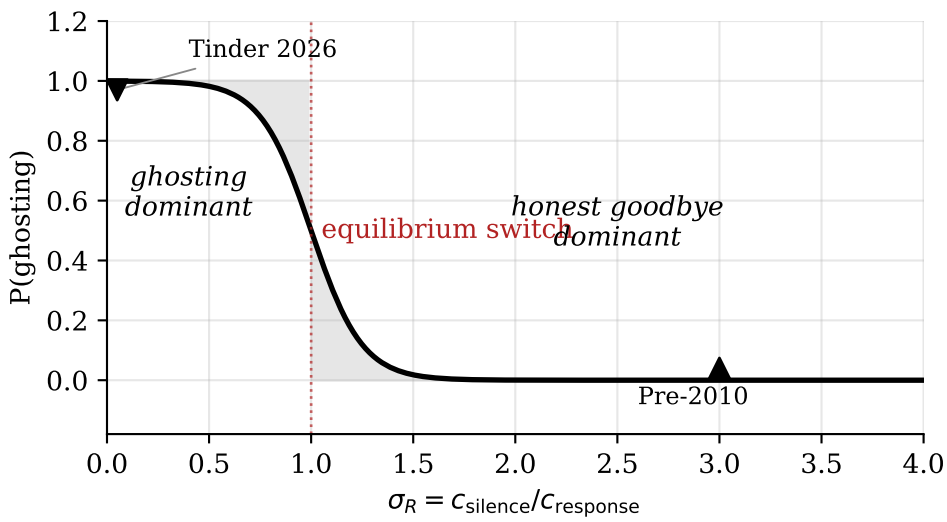
Theoretical model. Cost ratio values are structural estimates, not measured.

A match goes silent. The intuition is that the other person is afraid of confrontation, emotionally immature, or playing games. The mechanism is more boring. Dating-app architecture has driven the cost of silent exit below the cost of an honest goodbye. The Nash equilibrium has shifted. Most people are doing what the math tells them to do.



Setup. Define the *Silence-Cost-Ratio* $\sigma_R = c_{\text{silence}}/c_{\text{response}}$, where c_{silence} is the social-reputation cost of leaving a conversation without acknowledgment and c_{response} is the cost of writing an honest exit message. When $\sigma_R > 1$, honest goodbye is dominant. When $\sigma_R < 1$, ghosting is dominant. Pre-2010, $\sigma_R \approx 3$. Today on dating apps, $\sigma_R \approx 0$.

Definition 1 (Silence-Cost-Ratio)**.** *For a sequential dyadic game with exit option at each step,* $\sigma_R = c_{silence}/c_{response}$ *is the ratio of expected social cost of silent exit to the cost of an honest exit message. Both costs are functions of the recurrence probability* p *that the parties or their networks will encounter each other again.*

Theorem 1 (Equilibrium Switch)**.** *In a sequential dyadic exit game with finite horizon and discount factor* $\delta < 1$*:*

- *If* $\sigma_R > 1$*, the unique subgame-perfect equilibrium has both players sending honest exit messages when payoff falls below the continuation threshold.*
- *If* $\sigma_R < 1$*, the unique subgame-perfect equilibrium has both players ghosting at the first sign of declining payoff.*

The transition at $\sigma_R = 1$ *is sharp.*

Sketch. By backward induction. At the final exit decision, the dominated strategy is whichever has higher cost. By stage-game dominance, the same applies at every prior decision. The threshold is exactly $\sigma_R = 1$ where the costs equate. □

Why σ_R collapsed. The cost of being known as a ghoster depends on the recurrence probability p. In a small-network world (mutual friends, workplace, geographic proximity), p is high and reputation enforcement is real. In a continent-wide app pool, $p \to 0$, reputation memory disappears, and $c_{\text{silence}} \to 0$. Meanwhile c_{response} stays roughly constant: writing a goodbye message is no easier or harder than it ever was. The ratio collapses.

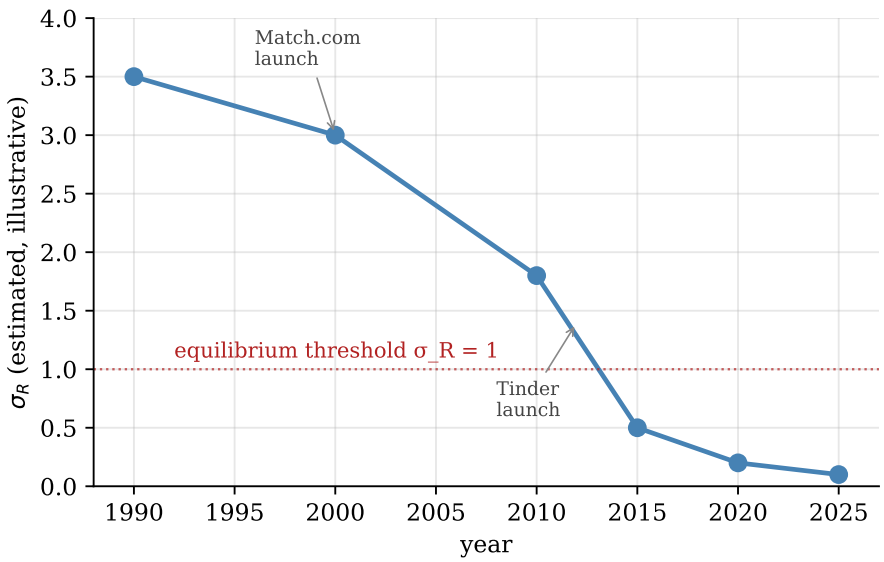
Corollary 1. *The shift in ghosting behavior between 2010 and 2026 is structural, not generational. Replacing the user pool with people from 1995 would produce the same equilibrium under the same platform architecture.*

Information-theoretic view. Communication consumes channel capacity. Honest exit messages carry $H_{\text{message}} \approx 5$–10 bits of information (acknowledgment plus reason). Ghosting transmits $H_{\text{silence}} = 0$ bits. The platform optimizes for total user retention, which is maximized when low-engagement conversations end with 0 bits of "negative" content rather than 5–10 bits of explicit rejection. The architecture rewards silent exit.

The Bridge.

$$\boxed{\sigma_R \;=\; f(p,\text{ platform architecture}) \;=\; \frac{c^{\text{base}}_{\text{silence}} \cdot p}{c^{\text{base}}_{\text{response}}}}$$

The Game-Theoretic equilibrium switch is driven by an Information-Theoretic property of the channel: anonymous platforms reduce p, which collapses c_{silence}, which flips the equilibrium.



Intervention pattern. The fix is not "be more emotionally mature." The fix is mechanism redesign that pushes σ_R back above 1. Hinge's "Your Turn" feature flags accounts that consistently leave conversations dangling and may down-rank them, adding an algorithmic c_{silence}. If the intervention works, σ_R rises and ghosting frequency drops in the affected cohort. This is empirically testable and falsifies the platform architecture hypothesis if no effect is observed.

Cross-domain pattern.

Domain	σ_R collapse trigger	Observed pattern
Dating apps	Anonymous matching ($p \to 0$)	Ghosting normalized
Restaurant reservations	Online booking, no shared community	No-show rate up
Job applications	Employer one-sided platforms	Application ghosting
Friendships in group chats	Silent exit with no acknowledgment	Friendship fade-outs

Empirical anchors.

Quantity	Value	Source
Pew Research dating-app survey	46% UK users describe experience "bad"	Pew 2024
Tinder UK user loss 2023–24	500,000 users (16% drop)	Mentor Resea
Male US users feeling "insecure"	64%	Mentor Resea
Restaurant no-show rate post-OpenTable	up $\sim$ 30% vs phone era	industry estin

References

[1] Akerlof, G. (1970). The market for lemons: Quality uncertainty and the market mechanism. *QJE* 84, 488–500.

[2] Fudenberg, D., Tirole, J. (1991). *Game Theory.* MIT Press.

[3] Mentor Research Institute (2026). Men are disillusioned with dating apps in the US and England.

[4] Pew Research Center (2024). Online dating in the United States.

[5] Tucker, C. (2019). Privacy, algorithms, and artificial intelligence. NBER Working Paper.

Notes on parameter estimates

The numerical values $\sigma_R \approx 3$ pre-2010 and $\sigma_R \approx 0$ on Tinder are illustrative structural estimates, not measured. The exact values depend on community size, mobility, and network density. The claim is the direction of change and the equilibrium threshold at $\sigma_R = 1$, both of which are mathematically exact under the model.

What the model does not explain: ghosting in long-term relationships (where p is high and the model predicts honest exit), individual variance within a fixed σ_R regime, and emotional or trauma-related avoidance (which operates on different timescales than rational equilibrium analysis). Those phenomena require additional mechanisms.